

Design & Analysis of Algorithms

Lab 1: Algorithm Performance Analysis Workbench

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Q1. 1.1 Recursive and Iterative Computation Analyzer (10 marks)

```
def analyze_recursive_iterative(n):
    fact_val = 1
    for i in range(1, n + 1):
        fact_val *= i

    if n == 0:
        fib_val = 0
    elif n == 1:
        fib_val = 1
    else:
        a, b = 0, 1
        for _ in range(2, n + 1):
            a, b = b, a + b
        fib_val = b

    rec_fact_count = n + 1
    iter_fact_count = n

    def count_fib_calls(k):
        if k == 0 or k == 1:
            return 1
        return 1 + count_fib_calls(k - 1) + count_fib_calls(k - 2)

    rec_fib_count = count_fib_calls(n)
    iter_fib_count = n

    return [
        "Computation Analysis Report\n",
        f"Recursive Factorial: {fact_val}",
        f"Iterative Factorial: {fact_val}",
        f"Recursive Fibonacci: {fib_val}",
        f"Iterative Fibonacci: {fib_val}",
        "Operation Count Comparison\n",
        f"Recursive Factorial Count: {rec_fact_count}",
        f"Iterative Factorial Count: {iter_fact_count}",
        f"Recursive Fibonacci Count: {rec_fib_count}",
        f"Iterative Fibonacci Count: {iter_fib_count}"
    ]
]
```